Procurement Management

Buying products and services from outside the team, from cloud subscriptions to outsourced development, and the contracts that govern them: deciding what to buy, choosing a seller, agreeing terms, and closing the deal cleanly.

PMBOK Guide 7th ed. **Case study: Realtime Quiz on Linode · gicitc.com**

AGENDA

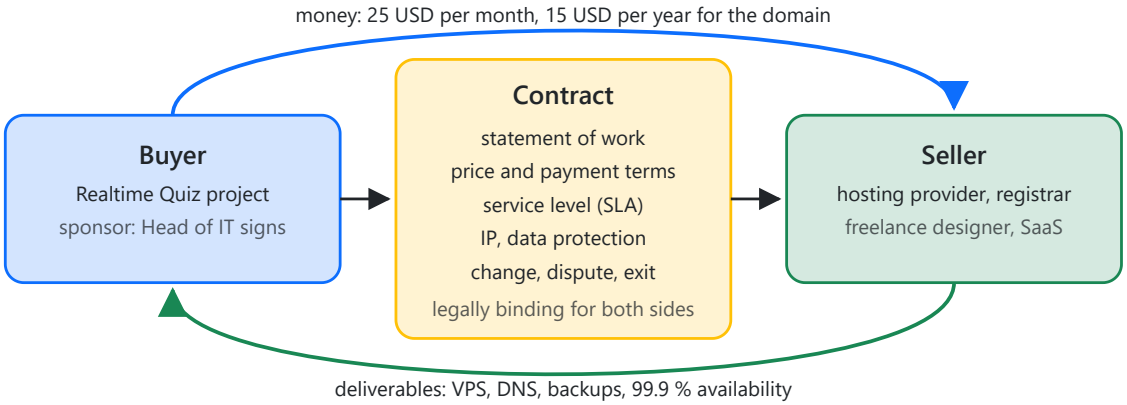
What we will cover

1. Make-or-buy analysis for IT components and services
2. Procurement management plan and the procurement statement of work
3. Contract types: fixed price, cost reimbursable, time and materials, and their risk allocation
4. Software licensing, SaaS subscriptions and cloud service agreements
5. Request for information, request for proposal, request for quotation
6. Source selection criteria and weighted evaluation of proposals
7. Negotiation basics and contract terms that matter in IT (SLA, IP, data protection, exit)
8. Contract administration: performance monitoring, change control, disputes
9. Outsourcing and offshore development: governance and communication
10. Closing procurements and lessons learned
11. Chapter Quiz
12. Lab-12

Click any item to jump directly to that topic.

Every IT project buys something

- **Procurement management** is the work of obtaining products and services from outside the project team, and of managing the agreements that come with them (PMBOK Guide 7th ed., Process Groups Practice Guide: plan, conduct, control procurements).
- Even a four-student project buys: a **Linode VPS** (about 25 USD per month), the **gicitc.com** domain, perhaps a freelance designer or a paid tool.
- A purchase creates a **contract**: a legally binding agreement in which the seller promises deliverables and the buyer promises payment. The project manager may not sign it, but must manage it.
- Buying moves risk, cost and control across the boundary: the chapter is about where that boundary should be and how to keep it under control.

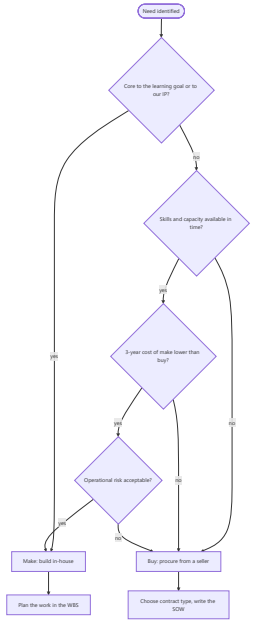


TOPIC 1

Make-or-buy: the first procurement decision

- **Make**: build or run it with our own people and assets. **Buy**: obtain it from a seller (product, subscription, service, staff).
- Compare on **total cost of ownership** over a realistic horizon (3 years for IT), never on the first invoice alone.
- **Risk factors**: lead time, availability, vendor lock-in, security and data location, dependency on one supplier.
- **Strategic factors**: is it core to our goal or IP? Do we want the skills in-house? For a student capstone the learning goal is the strategic factor.
- Output: a **make-or-buy decision memo** per item, with the numbers, the factors and the chosen option; buys feed the procurement plan.

> 💡 Capacity counts: 560 student hours in 14 weeks are already committed to the app; anything extra must be bought or dropped.



Hosting the Realtime Quiz: 3-year total cost of ownership

Hosting options, 3-year total cost of ownership (USD)

Row	Item		A Linode VPS	B Univ. server	C PaaS
2	Monthly fee		25	0	55
3	Months		36	36	36
4	Fees	=B2*B3	900	0	1,980
5	Domain gicitc.com	3 x 15	45	0	45
6	Setup effort		0	20 h x 30 = 600	0
7	Admin hours per month		1.5	2.0	0.5
8	Admin rate USD/h		20	30	20
9	Admin effort	=B7*B8*36	1,080	2,160	360
10	Total	=B4+B5+B6+B9	2,025	2,760	2,385

Rank by cost: A (2,025) < C (2,385) < B (2,760)

Option	Risk	Strategic fit
A Linode VPS (buy)	students administer the server; mitigated by daily backups	matches the SE course deployment: Docker Compose, Caddy, PostgreSQL
B University server (make)	6-week lead time, no SLA, holiday shutdowns, firewall exception needed	data on campus, but nothing learned about cloud operation
C PaaS (buy)	sticky sessions and Docker Compose limited; region fixed	least operations work, least learning

☑ **Decision: buy A.** Cheapest, fits the learning goal, and its main risk (operations by students) is exactly the skill the course wants to teach. The memo records B and C as fall-backs.

Build the quiz app or buy a quiz SaaS at per-student pricing?

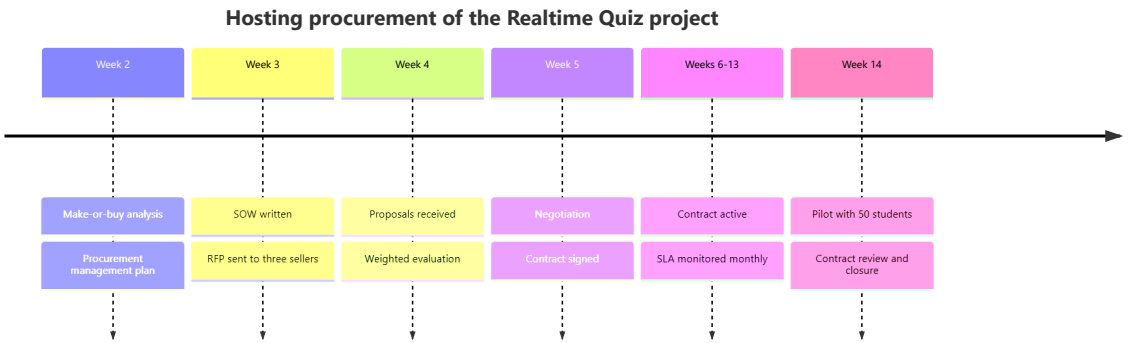
```
BUILD: our own Realtime Quiz             BUY: quiz SaaS, per-student plan
Effort 4 students x 10 h x 14 wk = 560 h  Price 3 USD / student / semester
Notional value 560 h x 15 USD  =  8,400   Pilot 50 students x 2 sem  =  300/yr
Hosting 3 years (previous slide) = 2,025  Department 600 students x 2 = 3,600/yr
Cash cost, 3 years            =  2,025    3 years, pilot scale        =    900
Full cost, 3 years            = 10,425    3 years, department scale   = 10,800

Break-even: 3 USD x 2 sem x N students x 3 yr = 10,425  ->  N = 579 / year
Cash view:  SaaS wins below 2,025 / 18 = 113 students per year
```

- **Two cost views**: cash only (what leaves the budget) and full cost (effort valued at a notional rate). State which one the memo uses.
- The 560 hours are **committed anyway** by the Software Engineering course: for this project their opportunity cost is zero, so the cash view applies.
- **Strategic factors**: the capstone is the learning goal; results stay in our PostgreSQL database; custom scoring and CSV export are required.
- **Risk factors**: SaaS price rises per student, data outside our control, features removed at the vendor's discretion.

Decision: make. Above 579 students per year the build is cheaper even at full cost; below it, the strategic factors decide.

From plan to close: the procurement process



Procurement management plan section	Realtime Quiz
Items to procure	VPS hosting, domain, optional designer
Contract type per item	subscription (monthly), FFP domain, T&M designer
Roles and authority	PM prepares, sponsor signs above 100 USD
Timetable and lead times	contract signed week 5, server live week 6
Constraints	university purchasing rules, cash budget from Lab-06
Risks (Lab-09)	hosting outage, vendor lock-in, data location
Metrics	on-time award, SLA met, spend vs budget

PMBOK Guide 7th ed.: plan procurements, conduct procurements (solicit, select, award), control procurements (monitor, change, close).

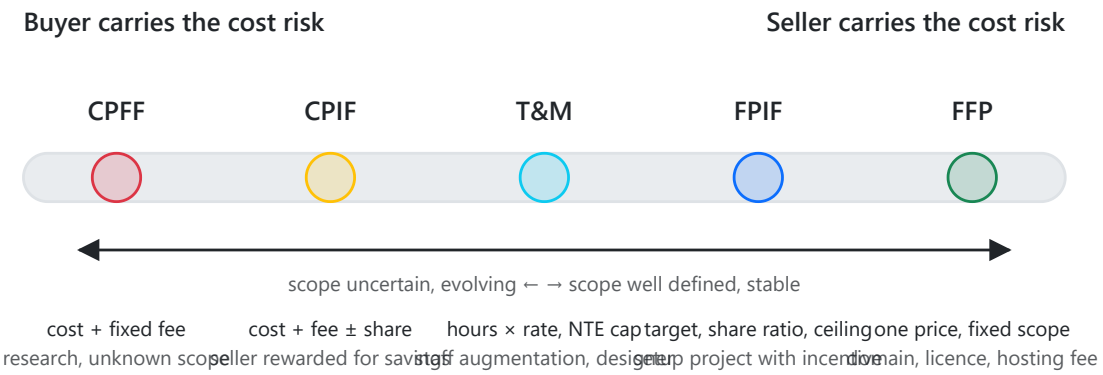
The procurement statement of work: what exactly are we buying?

```
# Procurement SOW: cloud hosting and domain
## 1. Background
Realtime Quiz (Spring Boot 4.1, PostgreSQL 17, Caddy reverse proxy),
pilot with 50 students in week 14, public address https://gicitc.com.
## 2. Scope of services
- Linux VPS: 2 vCPU, 4 GB RAM, 80 GB SSD, public IPv4, root access
- Docker Engine and Docker Compose v2 available on the image
- Daily automated backups, 7-day retention, restore on request
- Domain gicitc.com registered for 1 year, DNS A record to the VPS
## 3. Requirements
- Availability 99.9 % per calendar month, measured at the provider edge
- Data centre region: Europe; data and backups stay in the region
- Ports 80 and 443 open; WebSocket upgrade passes through
## 4. Deliverables and acceptance criteria
| Deliverable      | Acceptance test                                |
| VPS credentials  | ssh login works; docker compose version prints v2 |
| Domain live      | https://gicitc.com returns the login page, valid TLS |
| Backup restore   | one restore drill in week 5 passes             |
## 5. Period, price basis, reporting
Weeks 6 to 14 and 12 months after; monthly fee; cancel with 30 days notice;
monthly availability report by e-mail to the project manager.
```

- The **SOW** describes the part of the project scope that a seller delivers, in enough detail for sellers to price it and for us to accept it.
- Write **measurable** requirements: "99.9 % per month" and "restore drill passes", not "reliable" and "secure".
- Separate **scope** (what), **requirements** (how well), **deliverables** (evidence) and **acceptance** (the test we run).
- State the **period**, the **price basis** (monthly, fixed, hourly) and what the seller must **report**.
- The SOW becomes an attachment of the contract and the yardstick of contract administration.

⚠ A SOW that copies the seller's brochure buys whatever the seller sells, not what the project needs.

Contract types: who carries the cost risk?



PMBOK Guide 7th ed.: fixed-price, cost-reimbursable and time-and-materials families

Type	Buyer pays	Use when
FFP firm fixed price	agreed price, whatever it costs the seller	scope precise; seller prices in a risk margin
FPIF fixed price incentive fee	target price ± shared savings/overrun, capped by a ceiling	scope clear but cost uncertain; reward efficiency
CPFF cost plus fixed fee	all allowable costs + a fixed fee	scope unknown; buyer needs flexibility
CPIF cost plus incentive fee	costs + fee that moves with performance	uncertain scope, but measurable targets
T&M time and materials	hours × agreed rate + materials, usually with a not-to-exceed (NTE) cap	short, small, staff-like work; scope emerges

⚠ Fixed price is not "risk free" for the buyer: unclear scope turns into change orders at the seller's price.

Fixed price incentive fee: sharing savings, capping overruns




```
FPIF: seller sets up hosting and operates it for 12 months
Target cost   TC = 2,000     Target fee   TF = 200
Target price  TP = TC + TF = 2,200
Ceiling price CP = 2,480     Share ratio  80/20 (buyer/seller)

Point of total assumption:
PTA = (CP - TP) / buyer share + TC
    = (2,480 - 2,200) / 0.80 + 2,000 = 350 + 2,000 = 2,350

Actual 1,800: saving 200, fee = 200 + 0.20 x 200 = 240
              price = 1,800 + 240 = 2,040
Actual 2,300: overrun 300, fee = 200 - 0.20 x 300 = 140
              price = 2,300 + 140 = 2,440 (below ceiling)
Actual 2,600: above PTA, price = ceiling 2,480
              seller receives 2,480 for 2,600 of cost: loss 120
```

Below the PTA both share the outcome; above it the buyer's price is frozen at the ceiling and the seller has assumed the total cost risk.

TOPIC 4

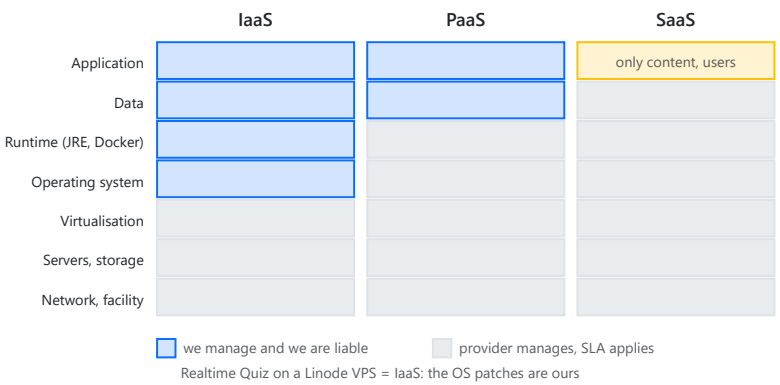
Licensing and SaaS models: what you pay for and what you may do

Model	You pay	Typical IT example	Watch for
Perpetual licence	once, plus 15 to 25 % per year maintenance	desktop IDE, on-premise database	upgrade rights end when maintenance lapses
Subscription per seat	per user per month or year	Jira Cloud, Microsoft 365	inactive seats still billed; annual lock-in
Usage-based	per GB, request, message	object storage, e-mail API, LLM tokens	runaway bills; set alerts and caps
Per learner / per student	per enrolled user per term	quiz SaaS, LMS plug-ins	cost scales with enrolment, not with value
IaaS compute	per hour or month per instance	Linode VPS 4 GB, about 25 USD per month	egress, backups and IPs billed separately
Open source	nothing, but obligations	Spring Boot and Caddy (Apache 2.0), PostgreSQL (PostgreSQL licence)	copyleft (GPL, AGPL) may oblige you to publish your source
Free tier / freemium	nothing until a limit	CI minutes, DNS, monitoring	limits change; migration cost when you outgrow it

- A **licence** grants the right to use software under conditions; you do not own the software. Read: number of users or devices, permitted purpose (education, commercial), term, transferability.
- A **SaaS subscription** bundles software, hosting and operations into one recurring fee: you buy a *service*, so the SLA and the exit clause replace the licence questions.
- A **cloud service agreement** is typically a click-through contract: non-negotiable for a small buyer, so the decision is "accept or choose another provider".
- Keep a **licence and subscription register** in the plan: what, who owns the account, renewal date, cost, exit plan.

```
Subscription register (Realtime Quiz)
Item        Owner      Renews      Cost/yr  Exit
Linode VPS  sponsor    monthly       300    snapshot + pg_dump
gicitc.com  lecturer   2027-09-15     15    transfer code
GitHub org  lecturer   free tier       0    git clone
```

IaaS, PaaS, SaaS: shared responsibility and the agreement review



Realtime Quiz on a Linode VPS = IaaS: the OS patches are ours

Cloud service agreement review, VPS provider for gicitc.com

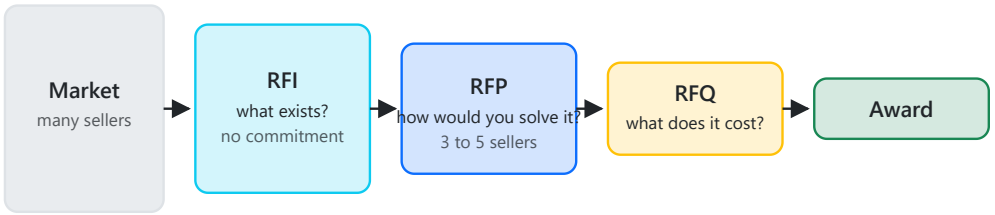
Clause	What the agreement says	Our action
Service level	99.9 % host and network	ask: credits, how measured
Data location	region chosen at creation	pick EU region, note in plan
Backups	paid add-on, daily, 7 days	include in price of option A
Price changes	30 days notice by e-mail	keep +10 % reserve (Lab-06)
Suspension	abuse or non-payment	sponsor's card, alerts on
Termination	any time, billed hourly	exit: snapshot + pg_dump
Liability cap	fees paid in the last 12 mo	our runbook covers the rest
Shared duties	OS, app, data are ours	patch schedule, TLS via Caddy
Governing law	provider's home country	accept: low-value contract

The review table goes into the procurement plan: it records what we could not negotiate and how the project compensates for it.

TOPIC 5

Asking the market: RFI, RFP, RFQ

each step narrows the field and sharpens the specification



small purchases skip steps: hosting for the Realtime Quiz needs an RFP only

	RFI	RFP	RFQ
Purpose	learn the market, shape requirements	get a proposed solution, team and price	get a price for a precise specification
Our input	problem statement, questions	SOW, evaluation criteria, contract terms	exact specification, quantities
Seller answer	capabilities, references, ideas	technical and management proposal, price	quotation, lead time, validity
Decided on	nothing yet	weighted criteria (technical + price)	mostly price
Realtime Quiz	which quiz SaaS exists, and at what pricing model	hosting service with SLA and backups	domain gicitc.com for 1 year

⚠ Fairness rules: same information to every seller, written questions and answers shared with all, a fixed deadline, criteria published before proposals arrive.

RFP excerpt: hosting for the Realtime Quiz platform

```
# Request for proposal RFP-2026-01: hosting for Realtime Quiz
Issued 2026-10-13 by the IT Department (buyer). Questions until 10-17,
answers published to all bidders 10-20, proposals due 10-24 17:00.

## 1. Purpose and background
Host a Spring Boot + PostgreSQL web application for live classroom
quizzes; pilot with 50 students in week 14; up to 2,000 WebSocket clients.
## 2. Scope of work            -> see attached SOW (2 pages)
## 3. Proposal contents (max 6 pages)
1. Technical: server size, region, backup, WebSocket and Docker support
2. Service: SLA, support hours, incident process, status page
3. Commercial: monthly price, setup fee, 3-year total, exit costs
4. References: two education or SME customers
## 4. Evaluation criteria and weights
| Criterion                          | Weight |
| Price (3-year total)               | 30     |
| SLA and availability history       | 20     |
| WebSocket, Docker Compose support  | 20     |
| Data location and security         | 15     |
| Support quality, documentation     | 10     |
| Exit and portability               | 5      |
## 5. Terms: monthly billing, 30-day notice, EU data location mandatory
```

- An RFP tells sellers **the problem, the constraints and how they will be judged**, and asks them to propose a solution.
- Publish the **weights**: sellers then invest in what matters to you, and the evaluation is defensible afterwards.
- Set a **page limit** and a **structure**: comparable proposals are easier to score.
- Run a **Q&A round**: questions from one bidder are answered to all, so no bidder has private information.
- State **mandatory terms** (data location, billing) up front: a proposal that fails them is not scored.

For a public university, procurement law may impose thresholds and forms; the RFP structure stays the same.

Source selection: the weighted evaluation matrix

Criterion	Weight	A Linode VPS	B University	C PaaS
Price, 3-year total	30	5.0	3.7	4.2
SLA and availability	20	4	2	5
WebSocket, Docker Compose	20	5	4	3
Data location and security	15	4	5	3
Support and documentation	10	4	3	4
Exit and portability	5	5	4	2
Weighted score (max 500)	**100**	**455**	**356**	**381**
Percentage and rank		**91 % · 1**	**71 % · 3**	**76 % · 2**

Scores 1 (poor) to 5 (excellent), agreed in a scoring guide before proposals are opened. Price is scored by formula, the other criteria by two evaluators independently, then reconciled.

```
Price score = 5 x lowest price / price
A: 5 x 2,025 / 2,025 = 5.0
C: 5 x 2,025 / 2,385 = 4.2
B: 5 x 2,025 / 2,760 = 3.7

Weighted score = SUM(weight x score)
A: 30x5.0 + 20x4 + 20x5 + 15x4
   + 10x4 + 5x5 = 455  ->  455/500 = 91 %

Spreadsheet (weights in B2:B7):
  C9: =SUMPRODUCT($B$2:$B$7,C2:C7)
  C10: =C9/(5*SUM($B$2:$B$7))
  C11: =RANK(C9,$C$9:$E$9)
```

☑ Recommendation: award to A. The matrix explains why to the sponsor and to the losing bidders.

Good criteria, gates, and a sensitivity check

Kind of criterion	Examples	How it is used
Gate (pass/fail)	EU data location, monthly billing, valid company registration	fail = proposal rejected before scoring
Technical	WebSocket support, backup restore time, capacity 2,000 connections	weighted 1 to 5 against the SOW
Management	support hours, incident process, named contact	weighted; evidence from the proposal
Past performance	references, availability history, status page	weighted; verify by calling a reference
Financial	3-year total cost including exit	formula score, never the only criterion

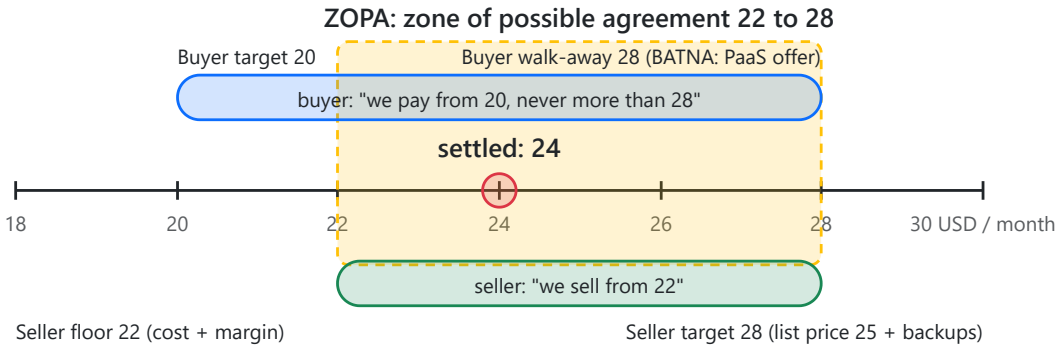
⚠ Common errors: weights that do not sum to 100, scoring price by gut feeling, changing weights after the proposals are opened, one evaluator only.

```
Sensitivity check: does the winner survive other weights?

Weights           base    price-heavy   quality-heavy
Price              30         45             15
SLA                20         20             25
WebSocket/Docker   20         15             25
Data, security     15         10             20
Support            10          5             10
Exit                5          5              5
                  ---        ---            ---
A Linode VPS      455        465            449  (rank 1, 1, 1)
B University      356        352            366  (rank 3, 3, 3)
C PaaS            381        394            370  (rank 2, 2, 2)

Ranking unchanged under all three weightings: robust award.
If two options swap places, discuss the criterion, not the weights.
```

Negotiation: BATNA, reservation price and the zone of agreement



trade: annual prepayment and a reference quote in exchange for 24 with backups included

- **Position**: what a party asks for (20 USD). **Interest**: why (a fixed cash budget, a working pilot in week 14). Negotiate interests, not positions.
- **BATNA**: best alternative to a negotiated agreement. Ours is the PaaS proposal; it sets our **reservation price** (walk-away), 28 USD.
- The **ZOPA** lies between the two reservation prices. If it is empty, stop and take the BATNA.
- **Trade variables** widen the zone: contract length, payment timing, backups, support level, a reference. Never trade away the SLA or the exit clause for price.
- Prepare a **negotiation sheet**: our target, our walk-away, their likely floor, three trades we can offer, one we will not.

Key IT contract clauses: SLA, IP, data protection, exit

Clause	What to write	Realtime Quiz example
Service level (SLA)	metric, target, measurement method, reporting, service credits, exclusions	99.9 % monthly availability at the provider edge; credits 10 to 50 % of the fee; planned maintenance announced 48 h ahead
Intellectual property	who owns deliverables, when ownership passes, licence-back for the seller's generic components, open source disclosure	designer's mock-ups and logo become university property on payment; app source stays ours; Apache 2.0 components listed
Data protection and location	roles (controller, processor), region, sub-processors, encryption, breach notification time, deletion	student names and scores are personal data; EU region only; breach notice within 72 h; deletion certificate at exit
Exit and data return	notice period, data export in open formats, transition assistance, deletion after return, no exit fee	30-day notice; full PostgreSQL dump and volume snapshot within 10 working days; deletion confirmed within 30 days
Liability and term	cap (often 12 months of fees), exclusions, term and renewal, price change rules	cap accepted; monthly term; 30 days notice of price change

```
7.2 Intellectual property
All designs, source code and documents
produced under this contract are made
for the University; ownership passes
to the University on payment of each
invoice. The Supplier keeps the right
to reuse generic, non-identifying
components.

9.1 Exit and data return
Within 10 working days after the end
date the Supplier delivers a complete
PostgreSQL dump and volume snapshot in
open formats, then deletes all copies
within 30 days and confirms in writing.
```

⚠ Clauses are drafted by the project, reviewed by legal or the purchasing office, signed by the sponsor. The PM never signs alone.

SLA metrics, service credits, and what 99.9 % really means

Metric	Target	Measured by	Penalty (service credit)
Availability per month	99.9 %	provider monitor at the edge, 1-minute probes	99.0 to 99.9 %: 10 %; 95 to 99 %: 25 %; below 95 %: 50 % of the monthly fee
Incident response (severity 1)	30 min	ticket timestamps	5 % per missed incident
Incident resolution (severity 1)	4 h	ticket timestamps	10 % per missed incident
Backup success	100 % of days	backup log, monthly report	restore drill at seller's cost
Planned maintenance notice	48 h	e-mail to PM	counts as downtime if not announced
Monthly report delivered	by the 5th	e-mail received	escalation, no credit

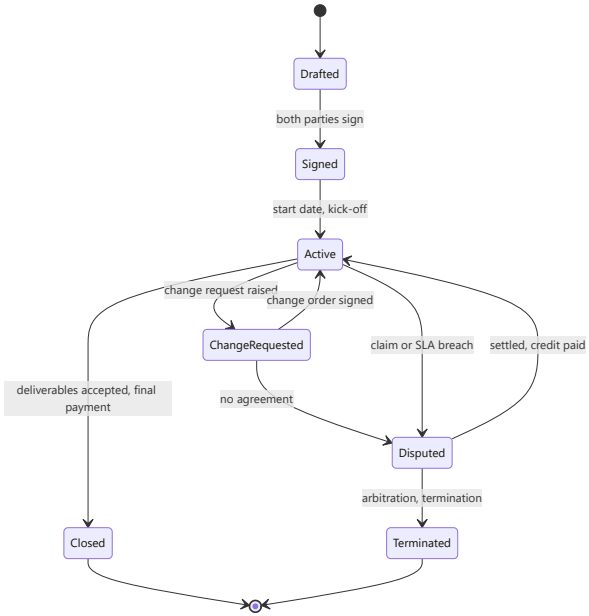
```
Availability = (minutes in month - downtime) / minutes in month
30-day month = 30 x 24 x 60 = 43,200 minutes
99.9 % allows 0.001 x 43,200 = 43.2 minutes of downtime
99.5 % allows 216 minutes;  99.0 % allows 432 minutes (7.2 h)

Incident: 3 h outage during the pilot evening = 180 minutes
Availability = (43,200 - 180) / 43,200 = 99.58 %
Credit band 99.0 to 99.9 % = 10 % of the fee = 0.10 x 25 = 2.50 USD

Lesson: a 2.50 USD credit does not repay a lost class session.
The SLA sets expectations; the project still needs a fallback:
docker compose up on the lecturer's laptop, local network join.
```

Credits are usually capped and must be claimed within a deadline: the monitoring plan (Topic 8) records every incident with start and end time.

Contract administration: the life of a contract



- **Monitor performance** against the SOW and SLA: monthly report, incident log, deliverable acceptance, invoice check.
- **Change control**: every change to scope, price or dates goes through a written **change order** signed by both sides, linked to the project change process (Lab-04).
- **Records**: contract, change orders, correspondence, acceptance notes, payments. They win disputes.
- **Relationship**: regular contact with a named seller counterpart prevents most disputes.

```
stateDiagram-v2
  [*] --> Drafted
  Drafted --> Signed : both parties sign
  Signed --> Active : start date, kick-off
  Active --> ChangeRequested : change request raised
  ChangeRequested --> Active : change order signed
  Active --> Disputed : claim or SLA breach
  Active --> Closed : accepted, final payment
```

Monitoring a T&M contract: the freelance UI designer

```
T&M contract: freelance UI designer for the Realtime Quiz
Rate 35 USD/h, estimate 20 h, not-to-exceed (NTE) 800 USD
Deliverables: style guide, 6 screen mock-ups, logo (SVG)
Weekly timesheet; written PM approval needed beyond 16 h
IP passes to the University on payment of each invoice

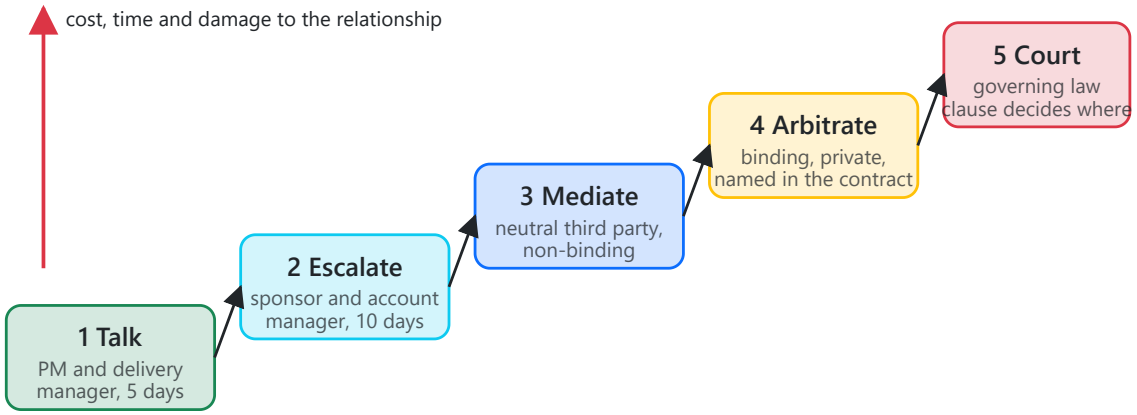
Week Hours Cum.h  Cost  Delivered        Status
 3      6     6    210  style guide      on track
 4      8    14    490  4 mock-ups       warn: 70 % hours, 57 % work
 5      6    20    700  6 mock-ups       approve +2 h for the logo
 6      2    22    770  logo             accepted, 30 USD under NTE

NTE check: 800 / 35 = 22.86 h  ->  at most 22 full hours billable
```

- **T&M risk**: the buyer pays for hours, not results. The **NTE cap** and an **approval threshold** convert open-ended cost into a bounded one.
- Monitor **burn versus delivery** every week: in week 4, 70 % of the hours bought 57 % of the work, so the PM asked for a re-estimate before approving more.
- **Acceptance** per deliverable, not per invoice: pay for the logo when the SVG passes the checklist (Lab-07 quality criteria).
- Small contracts still need a **contract summary sheet**: parties, dates, price basis, deliverables, SLA, contacts, change and exit clauses.

⚠ Verbal "just add another day" agreements are change orders without a record. Write a two-line change order instead.

Claims and disputes: agree the ladder before you need it

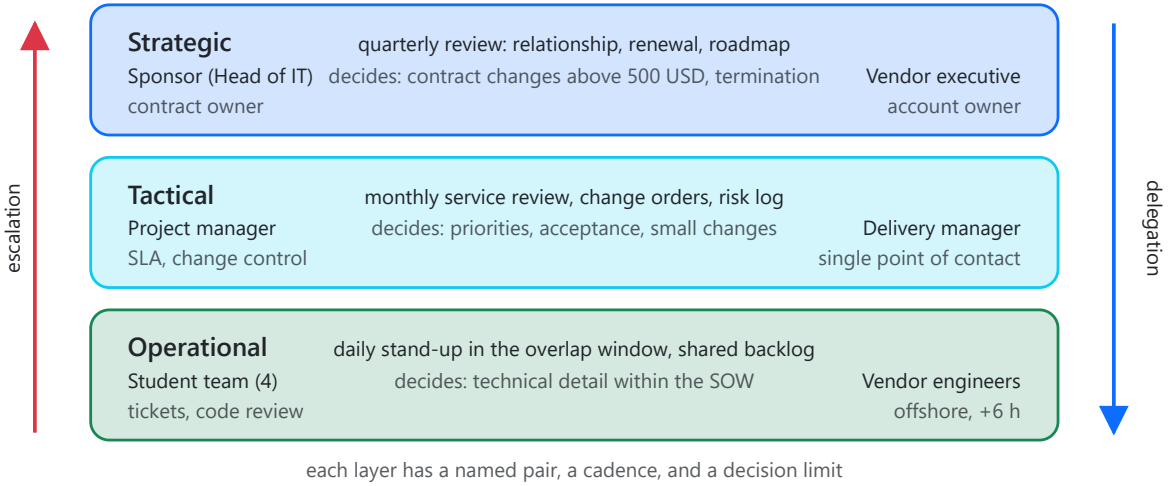


most IT disputes end at step 1 or 2 when the records are complete

Step when a claim arises	Evidence
1. Describe the breach against the SOW or SLA clause	clause number, dates, metric values
2. Quantify the impact and the remedy asked for	credit amount, redelivery, deadline
3. Notify in writing within the contractual deadline	e-mail with incident log attached
4. Negotiate; record the settlement as a change order	signed change order, credit on invoice
5. Escalate along the ladder only if unresolved	escalation log, dates of each step

Typical hosting claim: "3 h outage on 2026-12-09 19:00 to 22:00, availability 99.58 %, credit 10 % per clause 5.3": short, factual, sourced from the incident log.

Outsourcing and offshore development: governance in three layers



each layer has a named pair, a cadence, and a decision limit

- **Outsourcing** hands a whole function (development, operations, support) to a seller; **offshore** adds distance, time zones, language and legal systems.
- **Governance** answers: who talks to whom, how often, who decides what, and how problems climb. Three layers keep the sponsor out of daily detail and the team out of contract talk.
- **Communication** (Lab-10): fix an **overlap window**, written decisions, one shared backlog, demo every sprint, code in our repository from day one.
- **Knowledge and IP**: documentation as a deliverable, pair sessions, no single vendor person who knows everything.
- **Hidden costs**: management effort of 10 to 15 % of the contract value, travel, rework from misunderstood requirements.

Outsourcing the PDF and CSV export module to an offshore team

Forum	Participants	Cadence	Purpose and output
Kick-off workshop	PM, team lead, vendor DM, 2 engineers	once, week 7	walk through SOW, definition of done, repository access, coding rules
Daily stand-up	student on duty, vendor engineers	daily 09:15, 15 min, overlap window 09:00 to 11:00	blockers, ticket status in the shared GitHub Project
Code review	student reviewer, vendor author	per pull request, 24 h turnaround	merge only with passing CI and one approval
Sprint demo	team, PM, vendor DM	every Friday 10:00	working export shown on staging; acceptance notes
Service review	PM, vendor DM	monthly	hours vs NTE, quality metrics, change orders, risks
Steering	sponsor, vendor executive	end of contract	relationship review, lessons learned, reference

```
Contract: T&M, 2 engineers x 40 h, 25 USD/h
NTE 2,000 USD; approval beyond 60 h
Definition of done for every ticket:
  - code in our GitHub repo, CI green
  - unit tests, 80 % coverage on the module
  - reviewed by a student, merged by us
  - documented in README and UML (Ch. 7-12)
Time zone: vendor is +6 h
  their 15:00 to 17:00 = our 09:00 to 11:00
Hidden cost: PM effort 6 h/week x 4 weeks
  = 24 h, about 12 % of the contract
```

⚠ "They are cheaper per hour" ignores overlap constraints, review effort and rework. Compare total cost per accepted feature.

Closing a procurement: nothing is finished until the records say so

Closure item	Evidence	Owner	Realtime Quiz hosting
All deliverables accepted	signed acceptance notes	PM	VPS, domain, restore drill accepted weeks 6 and 5
Final invoice checked and paid	invoice, payment record	sponsor	monthly fees reconciled with the Lab-06 budget
SLA credits claimed and settled	incident log, credit note	PM	December outage: 2.50 USD credited
Data returned or deleted	export files, deletion certificate	team lead	pg_dump and snapshot archived; deletion confirmed
Credentials and access revoked or rotated	access review	team lead	ssh keys removed, JWT secret rotated, card removed
Licences and domain transferred	registrar record	lecturer	gicitc.com renewal owner named
Seller performance rated	rating in the vendor list	PM	4 of 5, would reuse
Lessons learned recorded	entry in the lessons register	PM	see template
Contract file archived	repository folder `procurement/closed/`	PM	contract, change orders, reports, e-mails

```
# Procurement lessons learned: hosting
## What worked
- Monthly billing kept the exit cheap
- Restore drill in week 5 found a
  missing volume in the backup set
## What did not
- SLA credit was trivial (2.50 USD)
- Domain renewal owner was not named
## Recommendations
- Name a renewal owner and set a
  calendar reminder 30 days ahead
- Keep a local fallback stack ready
- Publish RFP weights: it shortened
  the evaluation to two days
## Seller rating: 4/5, would reuse
```

Closing a procurement can happen long before the project closes (the designer in week 6) or after it (hosting runs 12 months). Close each contract on its own date.

WRAP-UP

Best practices and common mistakes

1 Decide with a 3-year TCO

Compare make and buy on total cost, risk and strategy over a realistic horizon, and write the memo even when the answer seems obvious.

2 Measurable SOW

Every requirement has a number and an acceptance test. "Reliable" is not a requirement; "99.9 % per month, restore drill passes" is.

3 Match the contract type to the uncertainty

Fixed price for clear scope, T&M with an NTE for small emerging work, incentives when you want the seller to save.

4 Publish weights, score with two people

Criteria and weights before proposals open, independent scoring, a sensitivity check, and a written recommendation.

5 Read the four clauses

SLA, IP, data protection and location, exit and data return. If you cannot negotiate them, plan around them and write that down.

6 Administer, then close

Log incidents, sign change orders, accept deliverables in writing, claim credits, return data, rate the seller, record lessons.

Mistake: buying the brochure

A SOW copied from the seller's website buys what they sell. Start from your requirements and the Lab-04 scope.

Mistake: price-only selection

The cheapest monthly fee with no backups, no exit and a fixed region costs the most when the pilot fails in week 14.

Mistake: verbal change orders

"Just add a day" without a written change order is an unbounded T&M contract. Two lines and two signatures are enough.

Chapter quiz 10 questions

1. In the hosting make-or-buy analysis, which factor is a *strategic* factor rather than a cost or risk factor?

- The 3-year total cost of 2,025 USD
- The 6-week lead time of the university server
- The deployment skills the students learn by operating the VPS
- The daily backup add-on price

2. FPIF: target cost 2,000, target fee 200, ceiling price 2,480, share ratio 80/20. What is the point of total assumption?

- 2,280
- 2,350
- 2,480
- 2,600

3. Two criteria weighted 60 and 40; a proposal scores 5 and 3 (scale 1 to 5). What is its weighted percentage?

- 80 %
- 84 %
- 90 %
- 96 %

4. An SLA promises 99.9 % availability per 30-day month. How much downtime is allowed?

- 4.3 minutes
- 43.2 minutes
- 432 minutes
- 7.2 hours

5. Which contract type leaves the buyer with the largest share of the cost risk?

- Firm fixed price (FFP)
- Fixed price incentive fee (FPIF)
- Time and materials with a not-to-exceed cap
- Cost plus fixed fee (CPFF)

6. When is a request for quotation (RFQ) the right instrument?

- When we do not yet know what the market offers
- When the specification is precise and price is the main criterion
- When we want sellers to propose a solution and a team
- When the contract is being closed

7. A T&M contract pays 35 USD per hour with a not-to-exceed cap of 800 USD. How many full hours can be billed at most?

- 20
- 22
- 23
- 25

8. Which clause guarantees that the project can leave the hosting provider with its PostgreSQL data in an open format?

- Liability cap
- Intellectual property
- Exit and data return
- Governing law

9. In the contract state diagram, a change request on an active contract is raised but the parties cannot agree. Which state follows?

- Closed
- Signed
- Disputed
- Drafted

10. Which activity does *not* belong to closing a procurement?

- Formal acceptance of all deliverables
- Returning or deleting the seller-held data
- Recording lessons learned and rating the seller
- Issuing a new RFP to the same sellers

WRAP-UP

Summary

Make-or-buy compares 3-year total cost, risk and strategic fit; the memo records the decision and the fall-backs.

The procurement plan says what to buy, with which contract type, who may sign, and when; the SOW says exactly what is bought and how it is accepted.

Contract types move cost risk from buyer (CPFF) to seller (FFP); T&M needs a not-to-exceed cap.

FPIF: PTA = (ceiling − target price) / buyer share + target cost; above it the seller carries every dollar.

Licences grant rights, subscriptions buy services, cloud agreements are accept-or-leave: keep a register and plan the exit.

RFI learns, RFP asks for a solution, RFQ asks for a price; publish criteria and weights before proposals open.

Weighted evaluation: SUMPRODUCT(weights, scores), a formula for price, two evaluators, a sensitivity check.

Know your BATNA and reservation price; trade term and payment, never the SLA, IP, data or exit clauses.

Administer the contract (monitor, change orders, claims) and close it: accept, pay, return data, rate the seller, record lessons.

Open Lab-12: 5 tasks + 1 challenge

Back to Chapter 01

Course wrap-up: with Lab-12 your project management plan for the Realtime Quiz is complete, from charter to procurement, twelve subsidiary plans in one repository.